

Where the Signature Is Born

Internal Emergence of Indefinite Signature from Null-Sector

Elimination in Resolution Geometry, and the Open Path to a Connection

Bridge

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Abstract

Resolution Geometry (RG) splits an observation system into a resolved sector R and a null sector N (the “hole”). This note reports a result that is entirely internal to RG and imports no physics: when the null sector is eliminated, it returns as a *memory kernel* (the Mori–Zwanzig projection, equivalently the Schur complement carried into the frequency domain), and the resulting effective “reactive form” $Q(\omega)$ changes inertia and grows a null cone. Three facts are established and numerically verified. (i) The emergent inertia marches through the *entire* staircase $(r, 0) \rightarrow (r - 1, 1) \rightarrow \dots \rightarrow (0, r)$ across resonance walls; a Lorentzian signature (one timelike direction) occurs, but only as *one band*, not as a forced or global structure. (ii) Once the form is indefinite a genuine, gauge-invariant, topological null cone appears. (iii) The signature structure is *not* a function of the static data (J, Σ) alone: two systems with identical (J, Σ) but different memory kernels have different signature walls, so connection reconstruction requires the triple (J, Σ, K) . We then state honestly what this is *not*—it selects no timelike direction, carries no Lorentz invariance, and touches neither curvature nor Einstein dynamics; at its strongest it lives in analogue-gravity territory, which reproduces Lorentzian kinematics but not dynamics. Finally we lay out the real open paths that would complete a QM–GR *connection bridge*—connection reconstruction, a holonomy–signature transport theorem, Einstein-compatible selection, and quantum-Fisher compatibility—each with a concrete, well-posed proposed finish. The claim of the paper is narrow and, we believe, defensible: the signature is born in the hole; the bridge is not yet built, but its remaining spans are now precisely specified.

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1 Scope

RG’s primitive is a local observation geometry $\mathcal{G} = (O, g; R, \Sigma)$ with $R = \text{Im}(J)$, $N = R^{\perp_g}$, and $g = \Sigma^{-1}$ positive-definite. Nothing in it is quantum or relativistic; it needs only observations, a measurement metric, and local response. Per the elimination results, physics is not built into $\mathcal{G}$ —it enters as dynamical laws placed *on top*. This note stays strictly inside that stance: every object below is RG-internal, and no claim about gravity or spacetime is made. What we ask is a structural question: *when the hole N is integrated out, what does it leave behind, and what does that force?*

2 The mechanism: the hole returns as a memory kernel

Let the joint (resolved $\oplus$ null) linear response have block generator with resolved mass/stiffness (M, K_R) , null dynamics A_N , and couplings A_{RN}, A_{NR} . Eliminating N is the projection-operator (Mori–Zwanzig) reduction; equivalently, it is the Schur complement taken in the frequency domain. It returns the null sector not as a static term but as a *memory kernel*

$$K(\tau) = A_{RN} e^{A_N \tau} A_{NR}, \quad \Sigma_R(\omega) = A_{RN} (i\omega I - A_N)^{-1} A_{NR},$$

and the effective (“reactive”) form on the resolved sector is

$$Q(\omega) = K_R - \omega^2 M - \text{Herm} \Sigma_R(\omega)$$

This $Q(\omega)$ is the object whose geometry we study. It is standard, established mathematics (dynamic stiffness / Mori–Zwanzig self-energy); the only RG-specific content is that N is precisely the *null sector of the observation geometry*, so the kernel is generated by eliminated *unresolved* directions, not by any spacetime input.

3 Established results

Lemma 1 (Signature staircase). *As ω increases through the resonance walls $\omega_i^2 = \lambda_i(M^{-1}K_R)$, the inertia (p, q) of $Q(\omega)$ steps through the full sequence*

$$(r, 0) \rightarrow (r - 1, 1) \rightarrow (r - 2, 2) \rightarrow \cdots \rightarrow (0, r).$$

By Sylvester’s law of inertia (p, q) is invariant under congruence (basis, coordinate, and gauge change), so each step is a genuine, gauge-invariant transition. The Lorentzian value $(r - 1, 1)$ is realized exactly on the band between the first and second walls.

Result 1 (Lorentzian is one band, not a selection). The staircase does not stop at one timelike direction. A Lorentzian signature appears, but as a single frequency band among r bands; nothing in the construction pins the system there, selects a unique time, or prefers $(r - 1, 1)$ over $(r - 2, 2)$. (*Verified: for $r = 4$, $(4, 0) \rightarrow (3, 1) \rightarrow (2, 2) \rightarrow (1, 3) \rightarrow (0, 4)$.*)

Result 2 (A gauge-invariant null cone). On any band with $q \geq 1$ the zero set $\{v : Q(v, v) = 0\}$ is a genuine null cone, topologically distinct from the point $\{0\}$ of a definite form. It cannot be removed by a smooth gauge transformation. (*Verified: a null direction with $Q(v, v) = 0$ to machine precision exists on the indefinite band.*) This is the internal geometric event “the hole grows a cone”—and it is generated by null-mode elimination, not inserted.

Result 3 (The kernel is required data). The signature structure is not determined by (J, Σ) alone. Two systems with identical static resolved data (M, K_R, A_{RN}, A_{NR}) but different null dynamics A_N —hence different memory kernels—have different signature walls. (*Verified: kernels*

$A_N = -\text{diag}(1, 2)$ and $-\text{diag}(0.4, 5)$ give 3 vs. 4 walls at different frequencies.) Consequently any reconstruction of a transport connection must take the triple:

$$(J, \Sigma, K) \longrightarrow [\nabla] \quad \text{---not---} \quad (J, \Sigma) \rightarrow \nabla.$$

Result 3 is the most useful outcome for the bridge program below: it identifies exactly the data a reconstruction theorem needs, and shows that the earlier hope of reconstructing a connection from the static pair alone was underdetermined.

4 The honest ceiling: what this is *not*

Remark 1 (Signature, not spacetime). Results 1–3 produce a *signature*, not a spacetime. The construction selects no timelike direction, carries no Lorentz invariance, imposes no causal ordering on the null cone, and—decisively—never touches a curvature tensor, a connection’s field equation, or a stress tensor. The chain $\det Q = 0 \Rightarrow \text{inertia jump} \Rightarrow \text{null cone is real}$, but it is a statement about an effective quadratic form, not about gravity.

Remark 2 (Analogue-gravity reference point, and its ceiling). The mechanism sits in the spirit of *analogue gravity*: perturbations on an underlying positive-definite medium can see an effective Lorentzian form (Unruh’s acoustic metric). That is a respected setting, and it is also the exact ceiling—analogue models reproduce Lorentzian *kinematics* (wave cones, horizon analogues) but provably *not* Einstein *dynamics*: the effective metric obeys no field equation. The present construction, at its strongest, inherits the same ceiling. This is why the result is reported as internal RG structure and not as a route to general relativity.

5 The open paths, and proposed finishes

The bridge is a program, not a theorem. What follows are the spans that remain, each now well-posed, with a concrete proposed finish. Dead routes (identifying $T_{\mu\nu}$ with J ; reading a bare determinant wall as global; treating the memory obstruction as a completed lock) are not listed—they are closed.

Open path 1 (Connection reconstruction $(J, \Sigma, K) \rightarrow [\nabla]$). *Goal*. Canonically build a connection (or a $\text{Aut}(\mathcal{G})$ -equivalence class) on the observation bundle from the triple. *Proposed finish*. The variation of $R_x = \text{Im}(J_x)$ already furnishes the canonical off-diagonal (second-fundamental-form) block: $A_{RN} = P_N(dP_R)P_R$, the Grassmann/projector connection of the resolved subbundle. Make it metric-compatible with $g = \Sigma^{-1}$ to fix the diagonal blocks ∇_R, ∇_N , and let the memory kernel K fix the reactive (frequency-dependent) part of transport. Prove this assignment is well defined modulo $\text{Aut}(\mathcal{G})$ and that its curvature reproduces the monodromy already constructed in the RG corpus. Each ingredient is established; the theorem is their compatibility.

Open path 2 (Holonomy–signature transport theorem). *Goal*. Show signature sectors are global (monodromy), not merely local walls. *Proposed finish*. Prove the reconstructed connection is Q -compatible ($\nabla Q = 0$ on each band), so holonomy $H_\gamma \in \text{O}(p, q)$. Then a loop that crosses a wall cannot be deformed to one that does not, because the structure group degenerates $\text{O}(p, q) \rightarrow \text{O}(p \mp 1, q \pm 1)$ through the singular form at the wall. This upgrades Result 1 from a local staircase to a genuine global obstruction, and connects it to the existing holonomy machinery.

Open path 3 (Quantum-Fisher compatibility (numerically testable now)). *Goal*. Decide whether QM’s own geometry induces the same transport as the RG reconstruction. *Proposed finish*. On a parametric quantum model, compute (a) the connection induced by the quantum Fisher / SLD metric and (b) the (J, Σ, K) -reconstructed RG connection, and test whether they

coincide or one projects onto the other. Unlike the others this is not only a theorem to prove but a computation to run; a positive result would secure the QM side of the bridge concretely.

Open path 4 (Einstein-compatible selection (hardest; stated honestly)). *Goal.* Characterize the subclass of admissible connections that obey an Einstein-like law. *Proposed finish.* Treat this as *selection*, not derivation: RG supplies the configuration space of admissible transports; a dynamical principle (e.g. an action on the space of connections) would pick the physical ones. We flag this as the least developed span and possibly out of reach with present tools; it should not be claimed until Paths 1–3 are in hand. Honest status: open, and not obviously closable.

6 Non-claims

Non-claim 1. No physics. The reactive form, staircase, and null cone are internal RG structure (dynamic stiffness / Mori–Zwanzig / Sylvester). Nothing here is a spacetime, a metric of gravity, or a stress tensor.

Non-claim 2. No emergent general relativity. A Lorentzian band is not Lorentz invariance, a causal cone, or Einstein dynamics; the strongest established cousin (analogue gravity) reaches Lorentzian kinematics only.

Non-claim 3. No completed bridge. Section 6 lists what remains; none of it is asserted as done. The single positive structural claim is Results 1–3: the signature is born in the hole, and the kernel is required data.

7 Conclusion

Eliminating the null sector does not erase it; it returns as a memory kernel whose effective form changes inertia and grows a null cone—so an indefinite, occasionally Lorentzian, signature is generated internally by RG, with no physics imported. That is real and, we think, elegant: the hole is where the signature is born. It is also bounded: it is a signature, not a spacetime, and it inherits the analogue-gravity ceiling. The value for the larger program is the sharpened target—connection reconstruction from (J, Σ, K) , a holonomy–signature theorem, and quantum-Fisher compatibility, with Einstein selection held open and honest. The bridge is not crossed; but the spans that remain are now named, well-posed, and, in one case, already testable.

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